

Study of Strong-Arm Latch as comparator and its application in ADC's

The Strong-arm latch is a circuit widely used in various applications, including as a sense amplifier, comparator, or robust latch etc. It offers advantages like Zero static power consumption, directly producing rail-to-rail output, input referred offset primarily arising from one differential pair, enhancing sensitivity. For this purpose, following paper was thoroughly studied and simulated in LTSPICE.

PAPER-1: Behzad Razavi, The Strong-ARM Latch, IEEE SOLID-STATE CIRCUITS MAGAZINE 2015, pp 12-17.

The working of the circuit can be summarised in following four phases:

- (i) Precharge phase where the inputs are disconnected, and internal nodes are charged to the supply voltage.
- (ii) Amplification Phase where input signals cause a differential current flow, which is amplified by the circuit.
- (iii) Regeneration Phase where the cross-coupled pairs of mosfets provide strong positive feedback, sharpening the difference between outputs.
- (iv) Output Stabilization Phase here the final rail-to-rail output is achieved through a feedback mechanism.

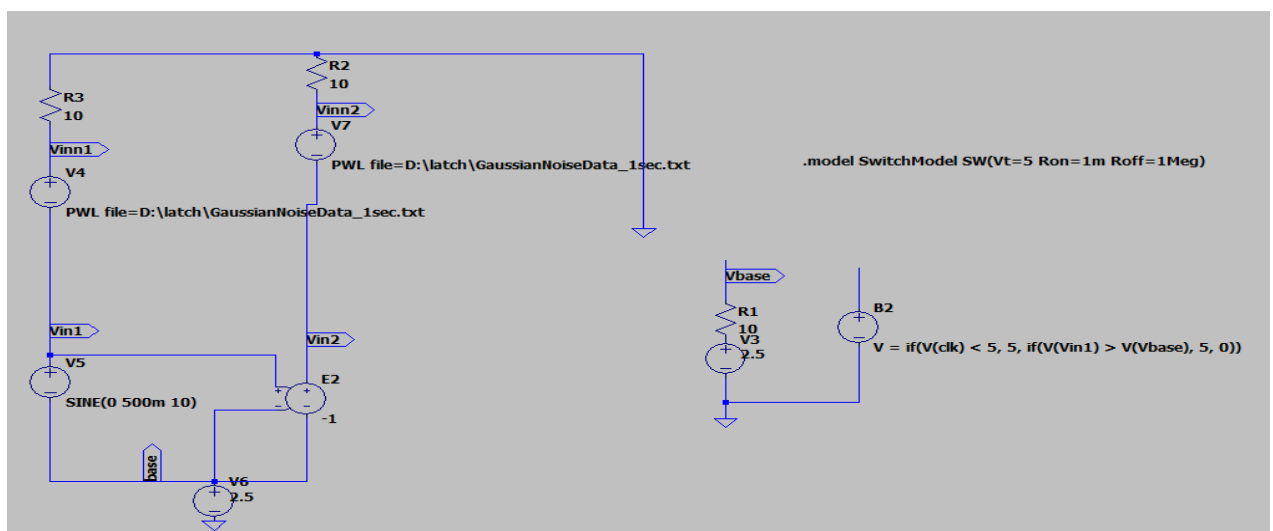
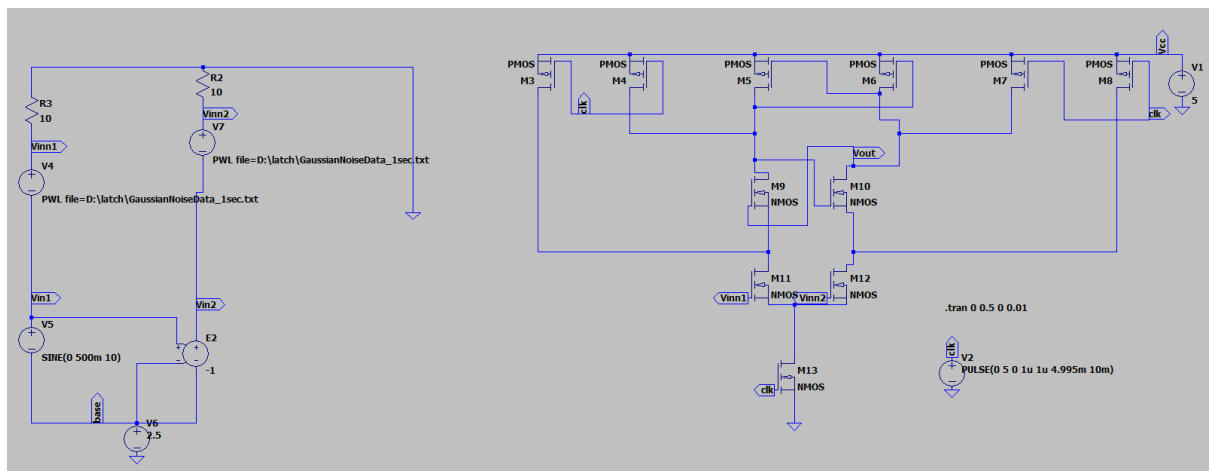
Challenges & Solutions: -

- i) Offset Voltage of the comparator- This was improved via capacitor mismatch adjustments or calibration circuits.
- ii) Electronic Noise- is a problem that starts from the main transistors. These noise levels can be managed by careful circuit design and transient analysis.
- iii) Kickback noise- This affects inputs due to feedback. This is minimized by clocking configurations and optimized component placement.
- iv) Supply Transients-caused by high peak currents during precharge. Hence, we adopt low supply impedance to avoid performance degradation.

SIMULATIONS

The circuit shown in Fig: 1 is the Strong-Arm Latch which acts as a comparator for various input signals. The circuit works with the condition that if clock is off then output is V_{cc} . If clock is on, then the circuit is acting as a comparator where the reference is set at 2.5v. So, signal magnitude above 2.5v give output of ($V_{cc}=5v$) and if below 2.5v then output is ($Gnd=0$). The graph Fig: 3 shows the input sinusoidal signal which is randomized using Gaussian noise (mean = 0, variance = 0.1). This noise created using MATLAB and then added through the PWL file addition in LTSPICE. This signal is cleaned of the noise using the differential amplifier. The signal is then used by the strong-arm latch comparator giving the output as shown in Fig: 5.

The ideal comparator in LTSPICE is a black box that gives output of 5v when clock is off and when clock is on according to signal magnitude above 2.5v give output of ($V_{cc}=5v$) and if below 2.5v then output is ($Gnd=0$). We then compared outputs of the comparator made with the Mosfet Circuit (Fig: 5) with an ideal comparator as shown in (Fig: 4). We observe that outputs both ideal comparator and Strong-Arm latch are almost identical. This shows the successful working of Strong-Arm Latch circuit.



LEGEND – X axis = Time, Y axis = Voltage

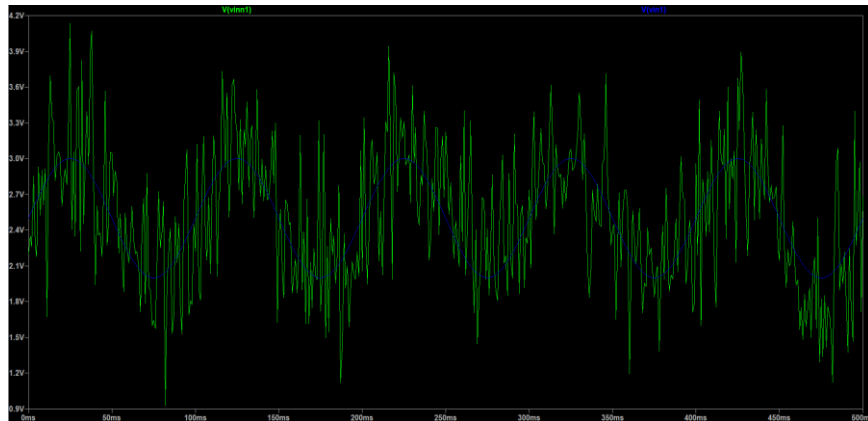


Fig 3: Input Signal

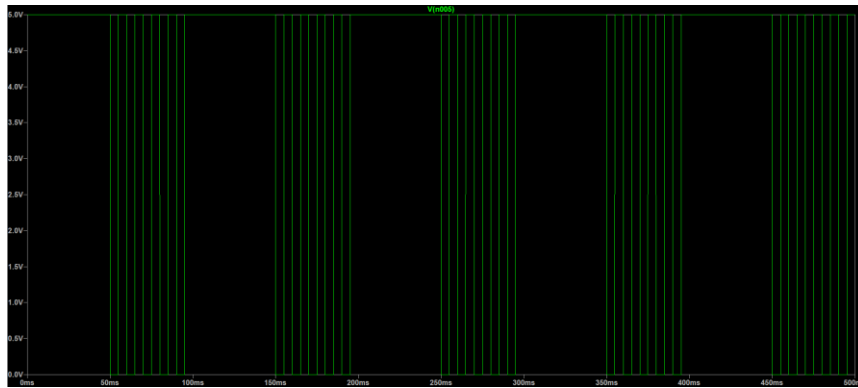


Fig 4: Output of Ideal Comparator

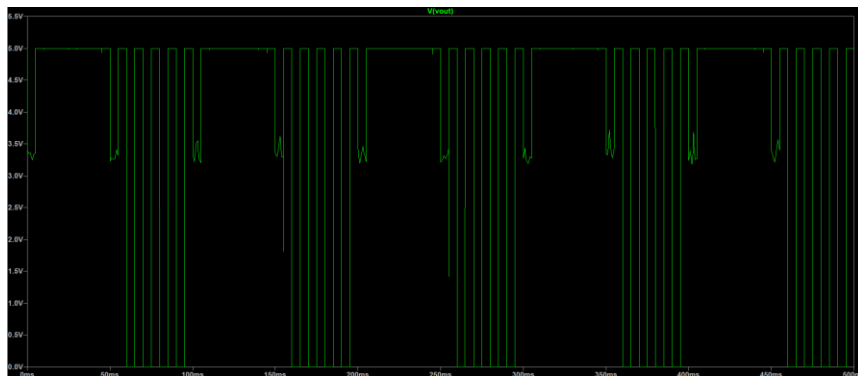


Fig 5: Output of Strong-Arm Latch

After completing simulations of paper 1 on Strong Arm Latch, the study of following paper was undertaken. PAPER 2 : (Venkata Srinivas, Shanthi Pavan, Ashish Lachhwani, and Naga Sasidhar, A Distortion Compensating Flash Analog-to-Digital Conversion Technique,

Some of the observations are listed below.

Flash ADCs are high-speed data converters that directly convert an analog signal to a digital output. The main components include a track-and-hold (T/H) circuit, comparator array, reference voltage ladder, and digital processing units. They find use in high-speed applications like disk drive read channels and wireless transceivers.

The main challenges faced in design of the ADC's are Nonlinearity in T/H Circuit referring to static nonlinearity in T/H which leads to distortion, limiting dynamic range and accuracy. Another challenge posed are the comparator offsets which are mismatches in comparators design which can degrade resolution.

The proposed technique was to predistort the reference voltages applied to comparators which compensates for the static nonlinearity in T/H, ensuring distortion-free operation. Another method was autozeroing technique which is used to correct comparator offsets in real-time. This adds bubble correction logic for continuous operation without interrupting the ADC's functionality.

"For the implementation of the technique predistortion was done to demonstrate to reduction of harmonic distortion effectively. Also, the comparator offset correction introduced very small signal degradation".

The designing was done by using a 0.35-micron CMOS which was designed for 6-bit resolution at 160 MSPS (Mega samples per second) with an ENOB (Effective Number of Bits) of up to 5.3 at Nyquist frequency.

The final efficacy of the method was revealed by Dynamic Testing which showed significant improvements in Signal-to-Noise and Distortion Ratio (SNDR) due to compensation techniques. There was reduced energy consumption by enabling a larger input signal swing while correcting for distortion. These are further validated through hardware tests, showing similarity with the simulations.